

# Relation Algebra

## §8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images:  $S(y) \subseteq R(x)$ . \ compares preimages:  $S^\circ(x) \subseteq R^\circ(y)$ .

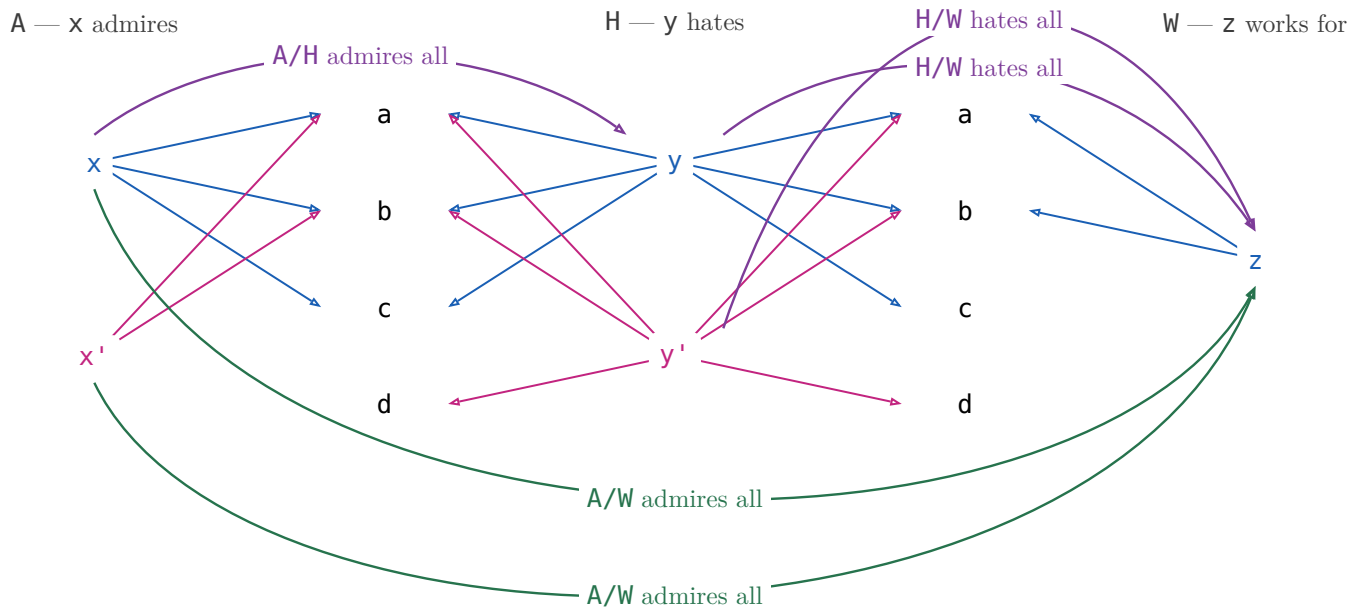
example: A admires, H hates, W works for

$x (A/H) y$  —  $x$  admires everyone  $y$  hates.

$x (H \setminus A) y$  — everyone who hates  $x$  admires  $y$ .

### §8.1. $(R/S)(S/W) \sqsubseteq R/W$

(8.1a)



(8.1b)

$x (A/H) y$  —  $x$  admires everyone  $y$  hates

$y (H/W) z$  —  $y$  hates everyone  $z$  works for

$x (A/W) z$  —  $x$  admires everyone  $z$  works for

$(A/H)(H/W)$  is a path:  $x \rightarrow y \rightarrow z$ , and that is all of it.  $A/W$  also holds of  $x'$ , who admires everyone  $z$  works for — but  $x'$  does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of  $(R/S)(S/W) \sqsubseteq R/W$ .

|                                                                                                                                                                                                                 |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $X \in R/S \iff XS \in R$<br>$X$ is any $x$ -to- $y$ pairing; one that only pairs $x$ with a $y$ such that $x$ admires everyone $y$ hates lies inside $R/S$ , and $R/S$ is the largest such.                    |  |
| $X \in S \setminus R \iff SX \in R$<br>The mirror — divide on the left when $x$ comes first.                                                                                                                    |  |
| $(R/S)S \in R$<br>There is a $y$ such that $x$ admires everyone $y$ hates, and $p$ is one of the people $y$ hates — then $x$ admires $p$ too. Strict at $S=\emptyset$ : $R/S$ is everyone, $(R/S)S=\emptyset$ . |  |
| $S \setminus (S \setminus R) \in R$<br>The mirror.                                                                                                                                                              |  |
| <b>associate:</b> $R/(S_1 S_2) = (R/S_2)/S_1$<br>A friend's enemy is two hops: divide by the far end first.                                                                                                     |  |
| $(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$<br>The mirror.                                                                                                                                        |  |
| <b>maps:</b> $f \setminus (R/S) = (fR)/S$<br>Rename $x$ before or after dividing — the licence to write $fR/S$ .                                                                                                |  |
| $R/(fS) = (R/S) f^\circ$<br>Rename $y$ : a map leaves a denominator as $f^\circ$ outside the box.                                                                                                               |  |
| $(R/S)(S/W) \in R/W$<br>Someone who admires all of a hate-set that already covers everyone $z$ works for admires those people too.                                                                              |  |
| $1 \in R/R$<br>$R/R$ runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only $a$ and $b$ admire each other's idols too, and still stay two people.                  |  |
| $(R/R)(R/R) = R/R$<br>$R/R$ is the preorder <b>admires at least as much as</b> , and a preorder is idempotent. Freyd writes $\sqsubseteq$ ; with $1 \in R/R$ above it is an equality.                           |  |
| $(R/R)R = R$<br>Reaching $p$ through someone whose idols $x$ fully admires is reaching $p$ directly, since $x$ admires their own idols.                                                                         |  |
| $R/1 = R$<br>Dividing by $1$ : $p$ 's set is just $\{p\}$ , so admiring all of it is admiring $p$ .                                                                                                             |  |
| $R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$<br>Admiring a combined hate-set is admiring each set in full.                                                                                                             |  |
| $S \setminus (R/W) = (S \setminus R)/W$<br>Which is why $S \setminus R/W$ needs no bracket.                                                                                                                     |  |

Fifteen laws, fifteen pictures, and not one shows a generator:  $\eta$ ,  $\upsilon$ ,  $\circ$  and composition are what the Frobenius generators build, and  $/$  is none of those — it is posited, with nothing to unfold.